

## Mark schemes

### Q1.

|     | Marking Instructions                                                                                                                   | AO     | Marks | Typical Solution                                                                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|--------|-------|-----------------------------------------------------------------------------------------------------------------------------|
| (a) | Reads graph and uses $10^{\log P}$ or $10^{\log V}$ to get either $P$ or $V$                                                           | AO3.4  | M1    | $\log_{10} P = 2.18$<br>$P = 151$<br>$\log_{10} V = -0.15$<br>$V = 0.708$                                                   |
|     | Correctly obtains $P$ and $V$<br>AWRT 150 and AWRT 0.71                                                                                | AO1.1b | A1    |                                                                                                                             |
| (b) | Calculates value of gradient to find $d$<br>Condone use of $\log d =$ gradient<br>Or uses a value of $c$ plus a $P/V$ pair to find $d$ | AO3.4  | M1    | $\log_{10} P = \log_{10} c + d \log_{10} V$<br><br>Gradient = $d = -1.4$<br><br>Intercept = $\log_{10} c$<br><br>$c = 93.3$ |
|     | Obtains correct value for $d$<br>AWRT -1.4<br>Not necessarily a decimal                                                                | AO1.1b | A1    |                                                                                                                             |
|     | Calculates value of intercept to find $\log_{10} c$<br>Or uses a value of $d$ plus a $P/V$ pair to find $c$                            | AO3.4  | M1    |                                                                                                                             |
|     | Calculates correct value for $c$<br>AWRT 93                                                                                            | AO1.1b | A1    |                                                                                                                             |
|     |                                                                                                                                        |        |       |                                                                                                                             |
| (c) | Uses their values of $c$ and $d$ in the formula $P = cV^d$                                                                             | AO1.1a | M1    | $P = 93.3 \times 2^{-1.4}$<br><br>$= 35.4$ kilopascals                                                                      |
|     | Obtains $P$ value, <b>including units</b><br>AWFW 30 to 40                                                                             | AO3.2a | A1    |                                                                                                                             |
|     |                                                                                                                                        |        |       | <b>Total 8 marks</b>                                                                                                        |

### Q2.

|     | Marking Instructions                                 | AO     | Marks | Typical Solution                             |
|-----|------------------------------------------------------|--------|-------|----------------------------------------------|
| (a) | Uses model to form one correct equation (PI by $a =$ | AO3.1b | M1    | $290 = a + b$<br>$331 = ae^{0.1} + be^{0.2}$ |

|                      |                                                                                                                                                                                                                                                                                                                                           |        |    |                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | 200) (ACF)<br>Accept $1.105a + 1.221b = 331$                                                                                                                                                                                                                                                                                              |        |    |                                                                                                                                                                                           |
|                      | Forms a second correct equation (ACF)                                                                                                                                                                                                                                                                                                     | AO1.1a | M1 | $290e^{0.1} = ae^{0.1} + be^{0.1}$<br>$b = \frac{(331 - 290e^{0.1})}{(e^{0.2} - e^{0.1})} = 90.3$<br>$= 90$ to the nearest integer<br>so $a = 200$                                        |
|                      | Obtains correct $a$ AWRT 200 and $b$ AWRT 90 (AG)<br>Only award if <b>both</b> previous M1's achieved. Do not award marks retrospectively for correct values of $a$ and $b$ used in part (b)                                                                                                                                              | AO1.1b | A1 |                                                                                                                                                                                           |
| (b)                  | Substitutes $t = 3$ and evaluates CAO                                                                                                                                                                                                                                                                                                     | AO3.4  | B1 |                                                                                                                                                                                           |
| (c)                  | Forms inequality (accept $<$ or $>$ ) (condone use of equation) FT 'their' value of $a$ , but $b$ must be 90                                                                                                                                                                                                                              | AO1.1a | M1 | $90e^{0.2t} > 200e^{0.1t}$<br>$e^{0.1t} > \frac{200}{90}$<br>$0.1t > \ln\left(\frac{200}{90}\right)$<br>$t > 10\ln\left(\frac{200}{90}\right) = 7.985$<br>Just less than 8 so during 2024 |
|                      | Uses logs or calculator to solve 'their' inequality (or equation)<br>If using trial and error must see $t = 7$ and $t = 8$ tested                                                                                                                                                                                                         | AO1.1a | M1 |                                                                                                                                                                                           |
|                      | Interprets final result. (Do not accept 2025)                                                                                                                                                                                                                                                                                             | AO3.2a | A1 |                                                                                                                                                                                           |
| (d)                  | Gives one limitation of the model.<br>Eg. Model must break down as both $n_A$ and $n_B$ will tend to infinity / model assumes nothing changes / no attempt to control the diseases / all the trees have died / finite number of trees / cure for the disease might be found / other factors such as drought could affect the model / etc. | AO3.5b | E1 | Eventually all of the trees will die so the model will no longer be accurate.                                                                                                             |
| <b>Total 8 marks</b> |                                                                                                                                                                                                                                                                                                                                           |        |    |                                                                                                                                                                                           |

**Q3.**

|     | Marking Instructions                                                                                                                                                                                                                                                                      | AO     | Marks | Typical Solution                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|-----------------------------------------------------------------------------------------------------|
| (a) | Uses model to form an equation to find $k$ with $t = 5.7$ , $m = \frac{1}{2}m_0$                                                                                                                                                                                                          | AO3.4  | M1    | $200 = 400e^{-k \times 5.7}$                                                                        |
|     | Obtains correct value of $k$                                                                                                                                                                                                                                                              | AO1.1b | A1    |                                                                                                     |
|     | Uses model to find $m$ with $t = 4$ , $m_0 = 400$ and <i>their</i> $k$<br>(Condone $m_0 = 200$ )                                                                                                                                                                                          | AO3.4  | M1    | $k = 0.1216047\dots$<br>$m = 400e^{-0.1216\dots \times 4}$                                          |
|     | Obtains correct value of $m$<br><b>CAO</b><br>(245.9296...)<br>AWRT 250                                                                                                                                                                                                                   | AO1.1b | A1    |                                                                                                     |
| (b) | Uses model to set up inequality or equation using <i>their</i> $k$ and 280                                                                                                                                                                                                                | AO3.1b | M1    | $400e^{-0.1216t} \leq 280$<br>$e^{-0.1216t} \leq 0.7$<br>$-0.1216t \leq \ln(0.7)$<br>$t \geq 2.933$ |
|     | Solves their inequality or equation to find $t$<br>(Follow through their $k$ only)<br>(2.933067)                                                                                                                                                                                          | AO1.1b | A1F   |                                                                                                     |
|     | Interprets <i>their</i> solution<br>(Only follow through if time is earlier than 1:42 pm)                                                                                                                                                                                                 | AO3.2a | A1F   |                                                                                                     |
| (c) | States any sensible reason such as:<br>Different people eliminate caffeine at different rates<br>The model is based on an average person<br>The length of time taken to drink two cups of coffee may have been significant<br>The amount of caffeine in a "strong cup of coffee" may vary | AO3.5b | B1    | Different people eliminate caffeine at different rates                                              |
|     |                                                                                                                                                                                                                                                                                           |        |       | <b>Total 8 marks</b>                                                                                |

**Q4.**

|      | Marking Instructions                                                         | AO     | Marks | Typical Solution                                                                                                                                                                                                                                                                                                      |
|------|------------------------------------------------------------------------------|--------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)  | States correct value CAO                                                     | AO3.4  | B1    | 50                                                                                                                                                                                                                                                                                                                    |
| (i)  |                                                                              |        |       |                                                                                                                                                                                                                                                                                                                       |
| (ii) | States correct integer value CAO                                             | AO3.4  | B1    | 609                                                                                                                                                                                                                                                                                                                   |
| (b)  | Forms correct equation and rearranges to obtain $e^{0.5t} = \dots$           | AO3.4  | M1    | $150 = 50e^{0.5t}$<br>so $e^{0.5t} = 3$                                                                                                                                                                                                                                                                               |
|      | Obtains the correct solution.<br>Must give answer to 3 sf                    | AO1.1b | A1    | $t = 2\ln 3 = 2.20$                                                                                                                                                                                                                                                                                                   |
| (c)  | 1 mark for any clear valid reason, must be set in context of the question    | AO3.5b | E1    | No constraint on the number of rabbits (ie could go off to infinity)<br><b>OR</b><br>Model is only based on the 3 years of the study. Things may change<br><b>OR</b><br>Continuous model but number of rabbits is discrete<br><b>OR</b><br>Ignores extraneous factors such as disease, predation, limited food supply |
| (d)  | Forms an equation with exponentials by letting $R = C$<br>PI                 | AO3.4  | M1    | $1000e^{0.1t} = 50e^{0.5t}$<br>$20 = e^{0.4t}$                                                                                                                                                                                                                                                                        |
|      | Solves 'their' equation correctly                                            | AO1.1a | M1    | $t = \ln 20 \div 0.4$<br>$= 7.49$                                                                                                                                                                                                                                                                                     |
|      | States correct answer as the year 2023 CAO<br>NMS scores full marks for 2023 | AO3.2a | A1    | 2023                                                                                                                                                                                                                                                                                                                  |
|      |                                                                              |        |       | <b>Total 8 marks</b>                                                                                                                                                                                                                                                                                                  |

**Q5.**

| Marking Instructions | AO | Marks | Typical Solution |
|----------------------|----|-------|------------------|
|----------------------|----|-------|------------------|

|                        |       |    |    |
|------------------------|-------|----|----|
| Circles correct answer | AO3.4 | B1 | 20 |
| <b>Total 1 mark</b>    |       |    |    |

**Q6.**

|      | Marking Instructions                                                                                                                                                                                           | AO     | Marks | Typical Solution                                                                                         |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|----------------------------------------------------------------------------------------------------------|
| (a)  | Translates proportionality into a differential equation involving $\frac{dA}{dt}$ , $A$ and a constant of proportionality.                                                                                     | AO3.3  | M1    | $\frac{dA}{dt} \propto A$ $\Rightarrow \frac{dA}{dt} = kA$ $\Rightarrow \int \frac{1}{A} dA = \int k dt$ |
|      | Separates variables                                                                                                                                                                                            | AO1.1a | M1    | $\Rightarrow \ln A = kt + c$ $\Rightarrow A = e^{kt+c}$                                                  |
|      | Integrates both of 'their' sides correctly                                                                                                                                                                     | AO1.1b | A1F   | $\Rightarrow A = Be^{kt} \quad \mathbf{AG}$                                                              |
|      | Constructs a rigorous mathematical argument that supports use of the given model.<br><b>AG</b><br>Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips. | AO2.1  | R1    |                                                                                                          |
| (b)  | States correct value of $B$                                                                                                                                                                                    | AO1.1b | B1    | $B = 0.25 \text{ or } B = \frac{1}{4}$                                                                   |
| (ii) | Uses $t = 20$ and $A = 0.5$ to find $k$                                                                                                                                                                        | AO3.1b | M1    | When $t = 20$ , $A = 0.5$<br>$\Rightarrow 0.5 = 0.25e^{20k}$                                             |
|      | Finds correct value of $k$                                                                                                                                                                                     | AO1.1b | A1    | $\Rightarrow 20k = \ln 2$ $\Rightarrow k = \frac{1}{20} \ln 2$                                           |
|      | Substitutes 'their' $k$ to get $A$ in terms of $t$                                                                                                                                                             | AO1.1a | M1    | $\Rightarrow A = \frac{1}{4} (e^{\ln 2})^{\frac{t}{20}}$                                                 |
|      | Constructs rigorous and convincing argument to show $A = 2^{\frac{t}{20}-2}$<br>Using correct notation throughout. <b>AG</b>                                                                                   | AO2.1  | R1    | $\Rightarrow A = 2^{-2} \times 2^{\frac{t}{20}}$ $A = 2^{\frac{t}{20}-2} \quad \mathbf{AG}$              |

|                       |                                                                                                                                      |        |    |                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|----|--------------------------------------------------------------------------------------------------------------------------------|
| (iii)                 | Uses the model to set up correct equation and attempt to find $t$                                                                    | AO3.4  | M1 | $2\pi = 2^{\frac{t}{20}-2}$<br>$t = 93.03$ days                                                                                |
|                       | Finds correct value of $t$                                                                                                           | AO1.1b | A1 |                                                                                                                                |
| (c)                   | States any sensible and relevant limitation of the model that is specified in terms of the pond, area, weed, rate of change or time. | AO3.5b | E1 | Model predicts that the area of weed will increase without limit and this is not possible since the area of the pond is $4\pi$ |
| (d)                   | Any sensible and relevant refinement to the model that is specified in terms of the pond, area, weed, rate of change or time         | AO3.5c | E1 | Introduce a limiting factor such as fish eating weed or rate of growth decreases as surface area covered                       |
| <b>Total 13 marks</b> |                                                                                                                                      |        |    |                                                                                                                                |

**Q7.**

(a) (i)  $t = 0 \quad N = 50$

B1

1

(ii)  $t = 24 \quad N = 345$

*Must be 345 (not 345.2534..)*

B1

1

(iii)  $1 + 9e^{\frac{t}{8}} = \frac{500}{400} \Rightarrow 9e^{\frac{t}{8}} = \frac{1}{4}$

*Correct algebra seen*

M1

$$e^{\frac{t}{8}} = 36$$

$$\text{Or } e^{-\frac{t}{8}} = \frac{1}{36}$$

m1

$$t = 8 \ln 36$$

$$\text{or } t = 16 \ln 6$$